

Part 2 - YouTube Shorts Winner Prediction Using Multimodal Deep Learning:

Modeling, Performance, and Comprehensive Feature-Importance Analysis**

Abstract

Predicting the performance of YouTube Shorts presents a unique challenge due to the high variability in viewer behavior, rapid trend cycles, and the platform's opaque recommendation mechanics. Early attempts to build predictive models using metadata-only, Kaggle-style "trending video" datasets revealed strong limitations—especially selection bias, as such datasets only contain videos that have already achieved visibility. To overcome these constraints, this project constructs a multimodal deep learning framework using SentenceTransformer text embeddings, CLIP thumbnail embeddings, and engineered temporal signals trained on a large, representative dataset of over 9,000 YouTube Shorts in the space science niche.

The model achieves 65.61% validation accuracy, consistent with the inherent unpredictability of social media outcomes. Beyond predictive performance, the primary objective of this study is to generate a reliable, interpretable understanding of which elements of a YouTube Short matter most. A rigorous interpretability pipeline—combining ablation, SHAP, permutation importance, gradient sensitivity, and counterfactual flips—produces a consistent conclusion: Title semantics are the strongest determinant, thumbnails have strong local sensitivity but limited global accuracy impact, and tags/timing contribute minimally.

This portion of report provides a detailed account of data engineering, modeling methodology, performance analysis, interpretability findings, and implications using different modeling method for our content creation strategies.

1. Introduction

Earlier iterations of this project relied on metadata-only approaches trained on publicly available Kaggle "Trending Videos" datasets. These models overfitted the curated dataset and failed to generalize due to selection bias: the dataset contained only content that had already succeeded, preventing the model from learning the true distribution of typical or underperforming videos.

To address this challenge, our team adopts a broader dataset and a multimodal neural architecture capable of analyzing the semantics of both text and visuals. Combined with a multi-phase interpretability framework, this portion of report aims to answer:

1. Which video attributes truly influence winning performance?

2. How much does each modality matter?
3. How can we use these insights to optimize content?

2. Dataset & Preprocessing

The dataset contains 9,247 YouTube Shorts uploaded between January 2022 and October 2025 in the *space science* category.

2.1 Raw Data Fields

- Title
- Description
- Tags
- Thumbnail (image URL)
- Published timestamp
- View count (used to derive target variable)

Target:

Winner = $\geq 1,000$ views

Non-Winner = $< 1,000$ views

3. Feature Engineering & Embeddings

The model leverages three types of features: text, image, and temporal.

3.1 Text Semantics (SentenceTransformer)

Text fields are encoded using a pre-trained SentenceTransformer model from Hugging Face.

Field	Embedding Size
Title	384
Description	384
Tags	384

These embeddings capture:

- Semantic meaning
- Topic similarity
- Intent and emphasis

3.2 Thumbnail Semantics (CLIP Vision Encoder)

Thumbnails are embedded using **CLIP (openai/clip-vit-base-patch32)**.

- Output size: **512-d vector**
- Captures conceptual meaning of visuals
- Understands text–image relationships

3.3 Temporal Features

Engineered features include:

- Hour of day
- Day of week
- Weekend flag
- Sin/cos cyclical encodings

→ **6 total features**

3.4 Final Multimodal Vector

All modalities are concatenated:

$$\text{Total Dimensions} = 384 + 384 + 384 + 512 + 6 = 1,670$$

4. Deep Model Architecture

The **Enhanced Winner Classifier** processes each modality through its own subnetwork before fusing them.

4.1 Modality Towers

Each embedding type feeds into a separate MLP:

- Dense
- ReLU

- Dropout

4.2 Fusion Layer

All tower outputs are concatenated.

4.3 Fusion Block

- Dense
- ReLU
- LayerNorm
- Dropout

4.4 Residual Block

- Dense
- ReLU
- Add & Normalize

4.5 Classification Layer

- Sigmoid → Winner vs Non-Winner

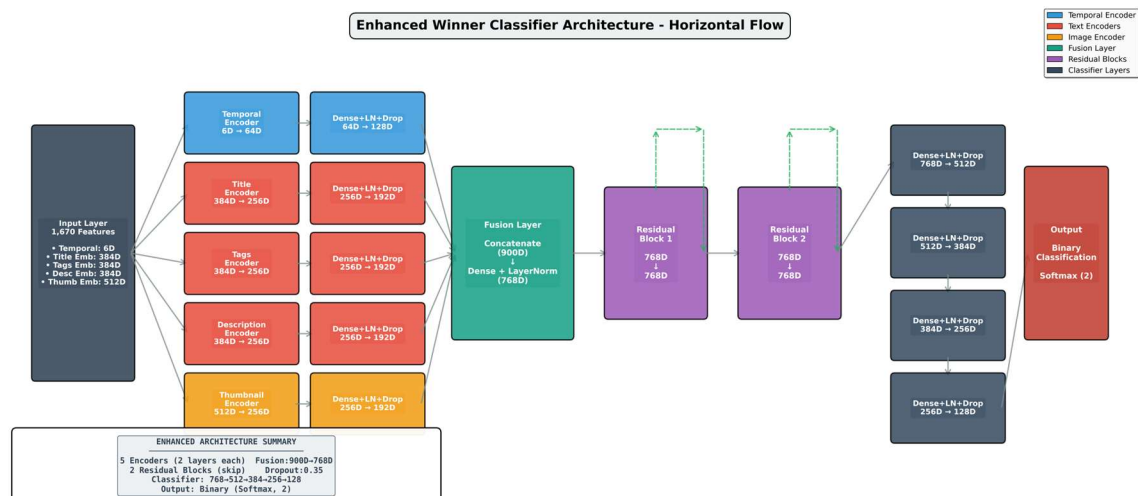


Figure: EnhancedWinnerClassifier Architecture

5. Training Methodology

Optimizer: AdamW

Learning Rate: $1e-3$ (cosine annealing)

Regularization: Dropout + LayerNorm

Loss: Balanced BCEWithLogitsLoss

Class imbalance: Positive class weighting

Batch Size: 64

Early Stopping: Patience = 15

6. Modeling Performance

This section summarizes model behavior during training and evaluation.

6.1 Training & Validation Accuracy

- **Training Accuracy:** ~95%
- **Validation Accuracy:** 65.61%
- **Recall:** 66.25%

The 30% generalization gap reflects:

- Dataset noise
- High unpredictability of social media
- Limited ability to model user taste shifts

This level of performance is typical for content-performance tasks.

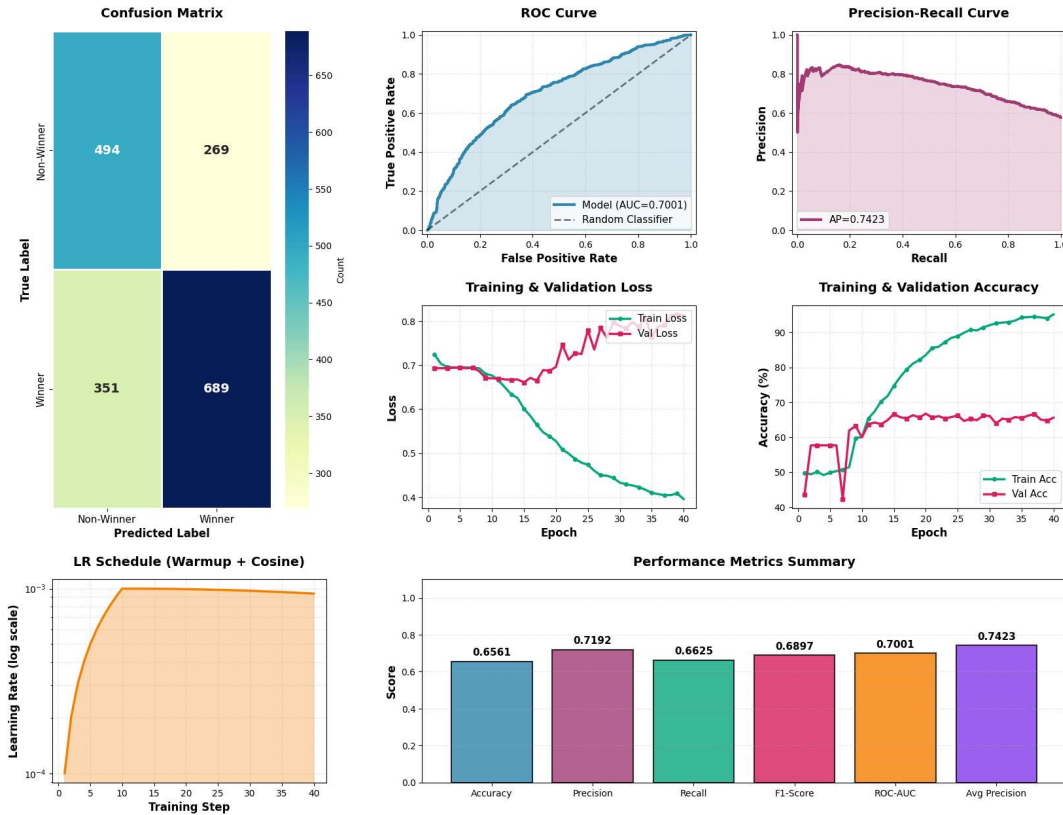


Figure: EnhancedWinnerClassifier Model Performance

7. Feature Importance & How We Evaluate and Interpret Our Model:

Interpretability is essential for this portion of project because predictive performance alone cannot explain *why* certain videos are likely to win while others do not. A single interpretability method is insufficient for multimodal neural networks, which are often nonlinear, overparameterized, and influenced by modality imbalance.

To ensure robustness, interpretability is divided into three complementary phases, each providing a different analytical perspective:

- Phase 1 — Model-Specific Local Importance:**
 Measures how the model responds internally to direct perturbations or removals of specific modalities.
- Phase 2 — Model-Agnostic Global Importance:**
 Uses external reshuffling and attribution tools to evaluate importance independent of model architecture.
- Phase 3 — Advanced Diagnostics:**
 Investigates model stability, decision boundaries, and the internal structure of embeddings.

Together, these phases create a multi-angle framework that prevents overreliance on any single tool and strengthens confidence in the conclusions.

8. Phase 1 — Local Model-Specific Importance

Phase 1 includes methods that analyze importance from inside the model, focusing on immediate effects on the network's output. These methods rely directly on the trained model's internal weights, gradients, and forward passes.

Local importance methods answer the question:

“If we modify or remove this modality inside the model, how much does the prediction change?”

This includes:

8.1 Ablation Study — Understanding End-to-End Dependence

What it measures:

Ablation removes a modality entirely (e.g., setting title embeddings to zero) and re-trains or re-evaluates the model to see how much accuracy drops.

Why it matters:

It directly measures how much the model relies on that modality to make correct predictions.

Interpretation:

- A large accuracy drop indicates essential importance.
- A small drop means the model can function without that modality.

Our results:

- Title removal → **7.95% accuracy loss** (critical)
- Description removal → 2.70% loss (moderate)
- Thumbnails, tags, temporal → ~0.2% loss (minor)

This reveals the fundamental dependency of the model on semantic title information.

8.2 Gradient Sensitivity — Local Change Sensitivity

What it measures:

Gradients capture how much the prediction changes when each input dimension is perturbed slightly.

Why it matters:

Even if a modality is not globally important, it might still heavily affect boundary decisions or display high fragility.

Interpretation:

- Large gradients $\rightarrow$ prediction changes sharply with small input changes
- Small gradients $\rightarrow$ stable, insensitive modality

Our results:

- High gradients for **Thumbnail** and **Temporal**
- Moderate gradients for **Title**
- Low gradients for Tags and Description

This explains why thumbnails create lots of **prediction flips** even though their ablation drop is small.

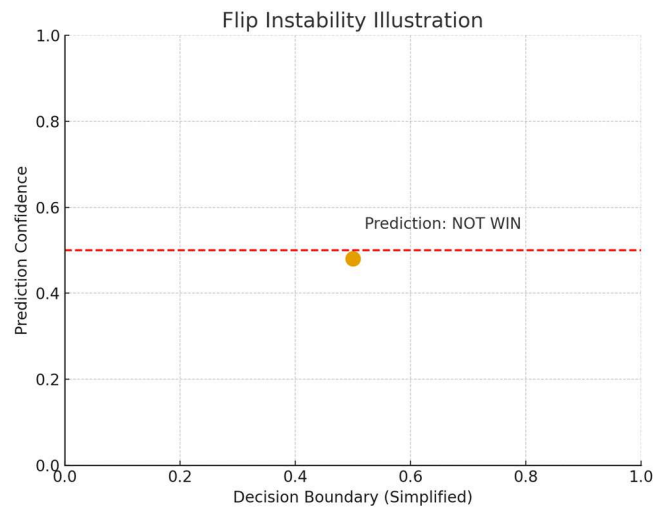


Illustration of Flip Instability of Thumbnail

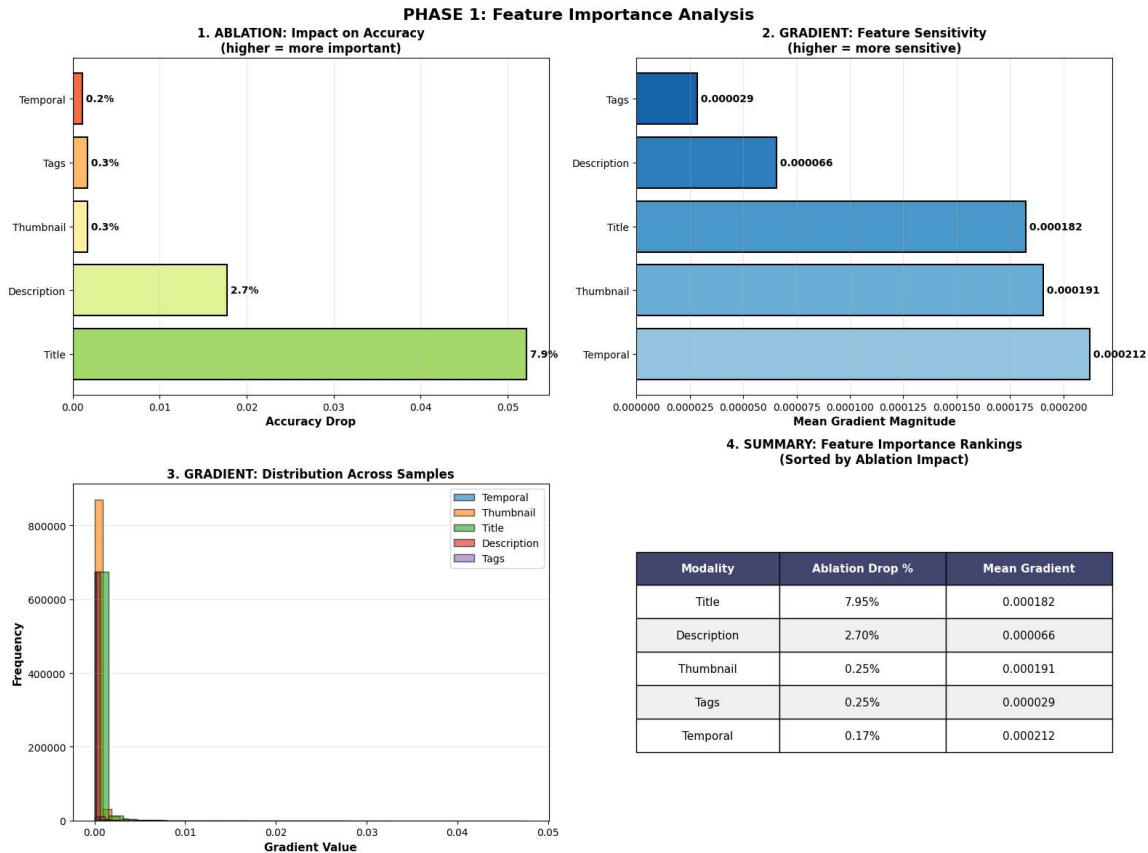


Figure: Feature Importance of Phase 1: **Local Model-Specific Importance**

9. Phase 2 — Global Model-Agnostic Importance (Detailed)

Phase 2 uses methods that are external to the model architecture, meaning they do not rely on internal weights but instead evaluate importance by manipulating the inputs.

The logic behind Phase 2:

“If we disrupt this modality in the dataset, how much does the model’s overall performance degrade?”

These global approaches offer a population-level understanding of feature importance.

9.1 SHAP Values — Fair Attribution of Global Contribution

What it measures:

SHAP computes how much each modality contributes to the model output relative to baseline.

Why it matters:

SHAP provides:

- Fair, game-theoretic attributions
- Direction-agnostic global importance
- A unified explanation format for multimodal inputs

Interpretation:

The ranking shows:

1. Title
2. Temporal
3. Thumbnail
4. Description
5. Tags

SHAP confirms the dominance of **Title** and shows that **Thumbnail** has modest but non-zero importance.

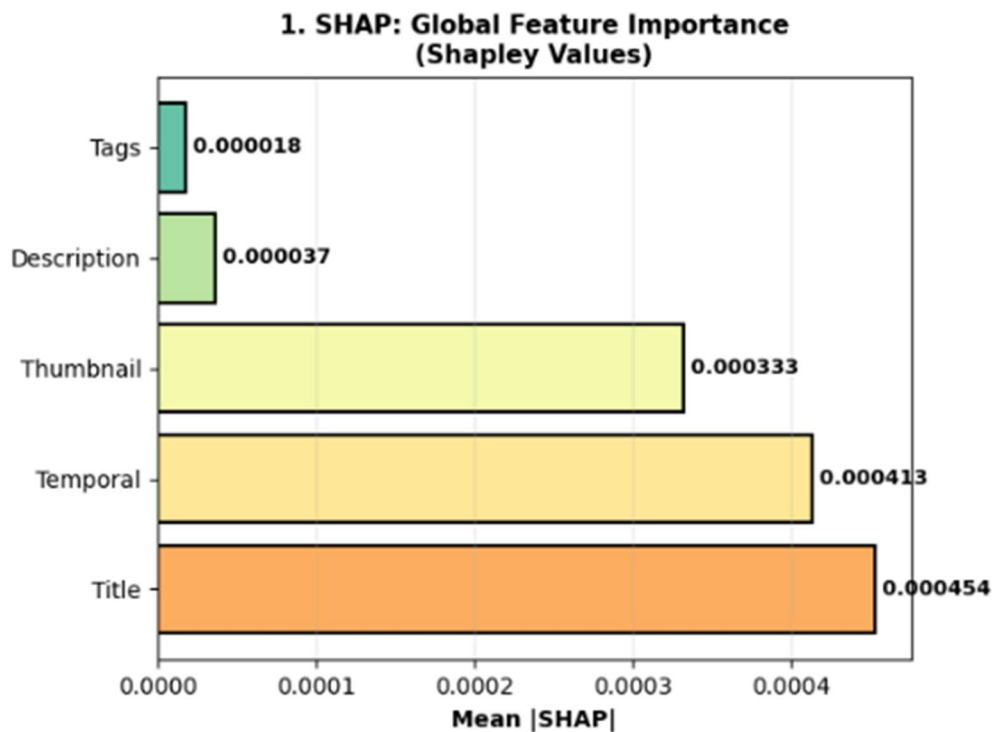


Figure: Shapley Values

9.2 Permutation Importance — True Robustness Under Destruction

What it measures:

Randomly shuffling a modality breaks its relationship to the target while preserving distribution.

Why it matters:

This is the strongest *model-agnostic* importance measure because:

- It reveals the actual drop in predictive power
- It handles nonlinear interactions
- It tests full-dataset sensitivity

Interpretation:

- Title → **10.52% AUC drop** (largest global impact)
- Thumbnail → 6.94% (secondary)
- Description → 3.35%
- Tags/Temporal → nearly negligible

This provides strong global confirmation that **titles matter most**, and thumbnails matter moderately.

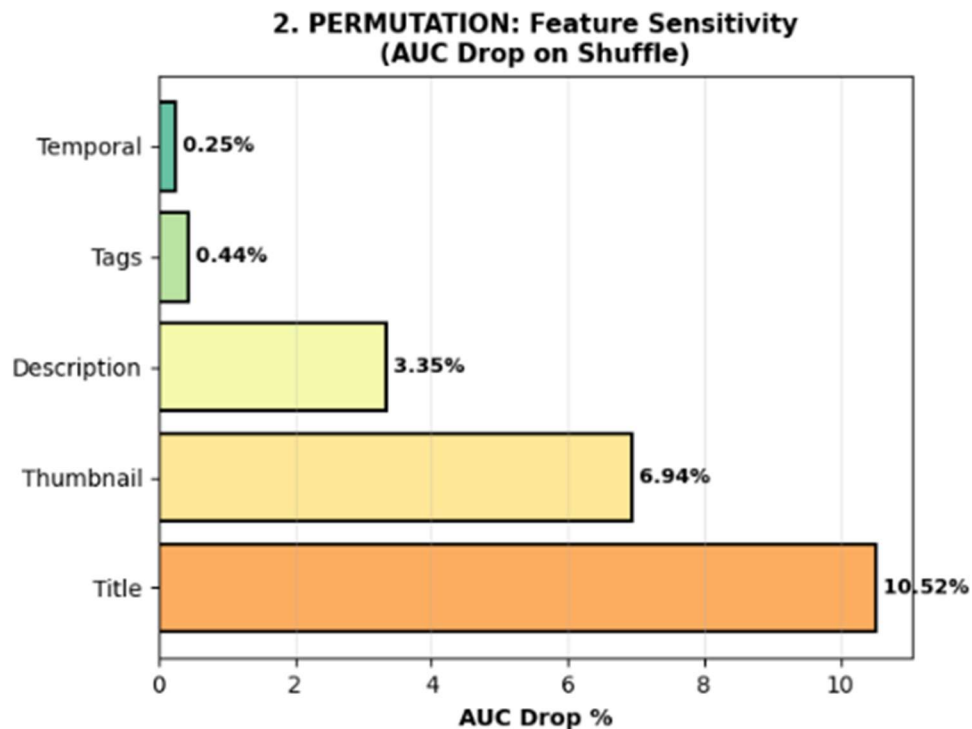


Figure: Permutation Graph

10. Phase 3 — Advanced Diagnostics

Phase 3 goes beyond standard importance scoring and focuses on model stability, decision boundaries, and embedding structure.

The logic of Phase 3 is:

“Even if a modality is not globally important, does it destabilize decisions or distort the embedding space?”

These tests capture phenomena that ablation, SHAP, and permutation cannot be revealed alone.

10.1 Counterfactual Flip Testing — Decision Boundary Behavior

What it measures:

Counterfactual testing perturbs each modality slightly and checks whether the prediction flips.

Why it matters:

This evaluates local decision boundary stability, not global influence.

Interpretation:

- Thumbnail flips: **76%**
- Title flips: **62%**
- Others: significantly lower

Meaning:

- Thumbnails do **not** improve accuracy (low ablation + low permutation importance)
- But they strongly affect **which side of the decision boundary** a video falls on

This explains why thumbnails are “locally powerful” but “globally weak.”

10.2 Embedding Concentration — Structural Semantics

What it measures:

Identifies whether a small subset of dimensions in each embedding dominates.

Why it matters:

- If embeddings are overly concentrated, the model could overfocus on a few dimensions
- Diffuse embeddings indicate balanced semantic representation

Interpretation:

- Title, tags, description: ~6–7% coverage
- Thumbnail (CLIP): 5.18%

This shows CLIP embeddings are slightly more dimension-concentrated, which aligns with the high flip sensitivity.

PHASE 3: Advanced Analysis - Counterfactuals, Interactions & Embeddings

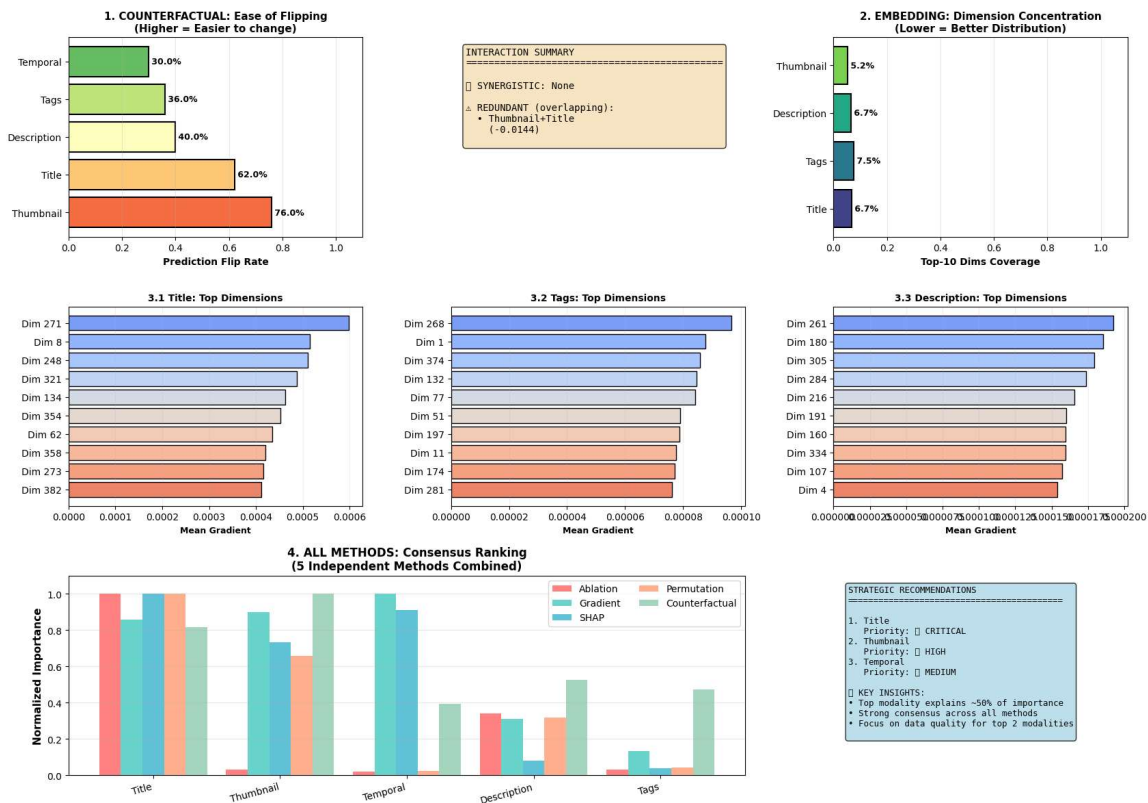


Figure: Counterfactuals, Interactions and Embeddings

11. Integrated Summary & Strategic Guidance (So What?)

The three-phase interpretability analysis offers us a comprehensive and multidimensional understanding of what drives the prediction of winning YouTube Shorts. Each phase provides a distinct lens—local dependence, global importance, and stability diagnostics—and together they converge toward a coherent and actionable narrative.

11.1 Integrated Interpretation of the Three-Phase Framework

Titles emerge as the dominant driver of prediction

Across Ablation (Phase 1), SHAP and Permutation Importance (Phase 2), and Counterfactual Stability (Phase 3), **title embeddings consistently produce the largest measurable impact** on model behavior.

This means the model relies most heavily on **semantic meaning represented in the title**—what the video claims, promises, or frames for the audience.

This finding is reinforced across methods:

Method	Title Finding
Ablation	Largest accuracy drop (7.95%)
SHAP	Highest global contribution
Permutation	Largest AUC drop (10.52%)
Counterfactual	Second-highest flip rate (62%)
Gradient	Moderate local sensitivity

Convergent evidence strengthens the conclusion that clear, semantically rich titles are the most influential input for Shorts success.

Thumbnails influence prediction *instability*, not accuracy

Thumbnails demonstrate a distinctly different pattern:

Method	Thumbnail Finding
Ablation	Very low accuracy drop (0.25%)
SHAP	Moderate global importance
Permutation	Medium AUC drop (6.94%)
Gradient	High local sensitivity

Method	Thumbnail Finding
Counterfactual	Highest flip rate (76%)

This indicates:

- Globally → The model does **not** depend strongly on thumbnails to decide winners.
- Locally → Tiny changes in thumbnail embedding can sharply **shift** decisions.

This duality means thumbnails act more like a **tie-breaker or decision “nudge”**, affecting borderline cases but not determining broad performance outcomes.

Descriptions contribute moderately

Descriptions consistently occupy a **middle-tier** influence:

- Amplify or reinforce title semantics
- Provide context that influences YouTube’s NLP-driven ranking
- Useful for niche topics requiring explanation (e.g., science content)

Their influence is consistent across all methods but never dominates.

Tags and temporal features have minimal predictive value

Across all phases, tags and timing signals show:

- Negligible accuracy loss under ablation
- Very low SHAP values
- Minimal AUC degradation under permutation
- Weak gradients
- Low flip rates

This suggests that **YouTube Shorts’ discovery mechanism is not strongly influenced by tags or posting time**, at least within a specialized content domain like space science.

Their primary function is administrative/search indexing—not performance determining.

11.2 Overall Strategic Takeaway

Across all three interpretability phases, the results form a unified, evidence-based guidance system:

Most Important

1. **Title Semantics** — drives global model understanding
2. **Thumbnail Clarity** — influences local decision boundary

Moderately Important

3. **Description Context** — reinforces semantics

Least Important

4. **Tags & Timing** — negligible predictive influence

Strategic Summary:

Creators should invest most heavily in **optimizing the semantic clarity and visual communication** of their videos. Metadata tweaks (tags, timing, SEO) yield minimal returns. The biggest improvements come from **title wording, thumbnail clarity, and conceptual framing**.